

Post-Hoc Tests with Tukey HSD

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Introduction

When a one-way ANOVA rejects equality of means across groups, post-hoc tests identify which specific pairs differ. Tukey's HSD (honestly significant difference) is the default pairwise comparison test after ANOVA with equal sample sizes, controlling the family-wise error rate at the nominal α .

Prerequisites

One-way ANOVA, multiple-comparisons problem.

Theory

For k groups with sample size n each and common residual $MS = MSE$, the Tukey critical value for pairwise differences is

$$q_{\alpha, k, n_{\text{total}} - k} \cdot \sqrt{MSE/n},$$

where q is the Studentised range quantile. Pairs whose observed mean difference exceeds this are declared significantly different at family-wise α .

Tukey's HSD is equivalent to a simultaneous test based on the maximum pairwise t statistic. Alternatives:

- **Dunnett**: comparing each treatment to a single control; more powerful than Tukey when this is the question.
- **Bonferroni**: simple but conservative; often over-adjusts.
- **Holm**: sequential, uniformly more powerful than Bonferroni.
- **Scheffe**: most conservative; valid for arbitrary contrasts, not just pairwise.

Assumptions

Same as ANOVA: independence, normal residuals, homogeneous variances. For unequal sample sizes, the Tukey-Kramer variant is used.

R Implementation

```
library(emmeans); library(multcomp)
set.seed(2026)

df <- data.frame(
  group = factor(rep(c("A", "B", "C", "D"), each = 30)),
  y      = c(rnorm(30, 50, 8),
             rnorm(30, 58, 8),
             rnorm(30, 60, 8),
             rnorm(30, 55, 8))
)
```

```
fit <- aov(y ~ group, data = df)

# Built-in Tukey HSD
TukeyHSD(fit)

# emmeans equivalent (more flexible, handles unbalanced designs)
emm <- emmeans(fit, ~ group)
pairs(emm, adjust = "tukey")

# Dunnett (comparing to A as control)
summary(glht(fit, linfct = mcp(group = "Dunnett")))
```

Output & Results

	diff	lwr	upr	p adj
B-A	8.421	3.02	13.82	0.0006
C-A	9.897	4.50	15.30	0.00005
D-A	4.975	-0.43	10.38	0.086
C-B	1.476	-3.92	6.88	0.892
D-B	-3.446	-8.85	1.96	0.353
D-C	-4.922	-10.32	0.48	0.090

Three of six pairwise comparisons are significant at family-wise $\alpha = 0.05$.

Interpretation

Report adjusted p-values and family-wise confidence intervals. “Tukey HSD post-hoc comparisons following a significant one-way ANOVA indicated that groups B and C each differed significantly from group A (both adjusted $p < 0.01$), while D did not differ significantly from A (adjusted $p = 0.09$).”

Practical Tips

- Only perform post-hoc tests after a significant omnibus F; otherwise you are fishing.
- For unequal sample sizes use Tukey-Kramer (R’s `TukeyHSD` and `emmeans` handle this automatically).
- Dunnett is more powerful than Tukey when all comparisons are to one control.
- For complex (non-pairwise) contrasts, use `emmeans` or `glht` with user-defined linear combinations.
- Conservative alternatives (Bonferroni, Holm) work regardless of equal variances; they are safer in heterogeneous-variance settings.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test

- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests